

Question	Marking Guidance	Mark	Comments
05.1	More than 1 <u>polypeptide</u> ;	1 (AO1)	Ignore prosthetic group OR named interactions between chains, eg disulfide bridge

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05.2	1. (Both) active sites have similar/identical tertiary structures OR (Both) active sites have identical amino acid sequences; 2. (So) form enzyme-substrate complexes (with the same substrate);	2 (2 x AO1)	1. Ignore shape for tertiary structure 1. Accept (both) have active sites that are complementary to different parts of the substrate; 1. Accept attach/bind for complementary 2 Accept E-S for enzyme-substrate

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05.3	1. Same volume of (each) buffer/pH solution; 2. Same concentration/mass of substrate (at start); 3. Same concentration/mass of denatured enzyme; If no marks gained, accept for 1 mark, Buffer and substrate and denatured enzyme OR Buffer and substrate and no enzyme OR Buffer and substrate and water;	3 (3 x AO3)	Ignore temperature Ignore amount for volume, concentration OR mass Accept pH solution for buffer 3. Accept description of denatured, eg boiled

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05.4	1. Both P and Q (are) active at pH 8.4; 2. P is (equally/most) active at both pHs OR Q is less active than P at both pHs OR P not affected by pH (change) OR Q is affected by pH (change); 3. Q is denatured/not active at pH 7.5 OR Q is less active than the control at pH 7.5; 4. Reaction occurs without enzyme(s) OR Reaction occurs in the control; 5. All (reactions) reach same end (point) OR Substrate is not used up;	3 max (3 x AO3)	Do not mark across all three conclusions Accept catalyse OR breakdown (substrate) OR hydrolyse (substrate) OR digest (substrate) OR cause reaction for active/reaction Ignore works better for active/reaction Ignore reference to optimum 3. Accept description of denatured 5. Accept broken down OR catalysed OR digested OR hydrolysed for used (up) 5. Accept does not reach zero for not used up

